



# VITAMINS: CLASSIFICATION, FUNCTIONS, AND HEALTH SIGNIFICANCE

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# CLASSIFICATION OF VITAMINS

- **Fat-Soluble (A, D, E, K):**

- Stored in liver and adipose tissue.
- Transported by chylomicrons.
- Risk of toxicity (Hypervitaminosis).

- **Water-Soluble (B-complex, C):**

- Not stored (except B<sub>12</sub>).
- Excreted in urine; low toxicity risk.
- Function primarily as coenzymes in energy metabolism.

A:



B1:



B2:



B3:



B5:



B6:



B8:



B9:



B12:



C:



D



E:



K:



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# FAT-SOLUBLE VITAMINS (VITAMIN A & D)

- Vitamin A (Retinoids):**

- Function:** Visual cycle (Rhodopsin), epithelial integrity, and gene expression.
- Deficiency:** Nyctalopia (Night blindness), Xerophthalmia, Bitot's spots.

- Vitamin D (Calciferol):**

- Function:** Calcium and phosphate homeostasis (acts as a hormone).
- Deficiency:** Rickets (children), Osteomalacia (adults).

# FAT-SOLUBLE VITAMINS (VITAMIN E & K)

- **Vitamin E (Tocopherol):**

- **Function:** Potent antioxidant; prevents lipid peroxidation in cell membranes.
- **Significance:** Protects RBCs from hemolysis.

- **Vitamin K (Phylloquinones/Menaquinones):**

- **Function:** Post-translational modification (gamma-carboxylation) of Clotting Factors II, VII, IX, and X.
- **Clinical:** Warfarin antagonism; Hemorrhagic disease of the newborn.

# SOURCES OF FAT-SOLUBLE VITAMINS

## Vitamin A:

- *Pre-formed (Retinols)*: Liver, kidney, egg yolk, whole milk, and butter.
- *Pro-vitamin (Carotenoids)*: Yellow and dark green vegetables (carrots, spinach), pumpkins, and papayas.

## Vitamin D:

- *Endogenous*: Synthesis in the skin via UV light (7-dehydrocholesterol).
- *Dietary*: Fatty fish (salmon, mackerel), fish liver oils (Cod liver oil), and fortified milk/cereals.

## Vitamin E:

- Vegetable oils (wheat germ oil is the richest), sunflower seeds, nuts, and green leafy vegetables.

## Vitamin K:

- *K1 (Phylloquinone)*: Green leafy vegetables (cabbage, kale, spinach).
- *K2 (Menaquinone)*: Synthesized by **intestinal bacterial flora**; also found in fermented foods.



# FAT SOLUBLE VITAMINS

**A**

Fat Soluble



Vision,  
Reproduction,  
Bone Health,  
Immune System,  
Skin

**D**

Fat Soluble



Strengthens Bones,  
Calcium Absorption,  
Immune System

**E**

Fat Soluble



Immune System,  
Flushes Toxins

**K**

Fat Soluble



Blood Clotting,  
Bone Health

# WATER-SOLUBLE VITAMINS: THE ENERGY RELEASERS

- **B1 (Thiamine)**: Coenzyme for TPP (Pyruvate Dehydrogenase).  
*Deficiency: Beriberi, Wernicke-Korsakoff.*
- **B2 (Riboflavin)**: Precursor to FAD/FMN.
- **B3 (Niacin)**: Precursor to NAD/NADP.  
*Deficiency: Pellagra (Dermatitis, Diarrhea, Dementia, Death).*
- **B5 (Pantothenic Acid)**: Component of Coenzyme A.

- **B1 (Thiamine):** \* Requirement increases with high-carbohydrate diets.  
    **Dry Beriberi:** Peripheral neuropathy and muscle wasting.  
    **Wet Beriberi:** High-output cardiac failure and edema.
- **B2 (Riboflavin):** \* FMN and FAD are prosthetic groups for oxidation-reduction enzymes (e.g., Succinate Dehydrogenase).
- **Clinical:** Cheilosis (cracks at corners of mouth) and glossitis (magenta tongue).
- **B3 (Niacin):** \* Synthesized from **Tryptophan** (requires B6).  
    **Hartnup Disease:** Deficiency due to impaired tryptophan absorption.



# Vitamin B Complex

| Vitamins                                                                                                                   | Functions                                                          |
|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|  <b>Vitamin B1<br/>(Thiamin)</b>          | Helps turn carbohydrates into the energy our bodies need.          |
|  <b>Vitamin B2<br/>(Riboflavin)</b>       | It helps boost immune system and also help treat headaches.        |
|  <b>Vitamin B3<br/>(Niacin)</b>           | Lowers harmful cholesterol while increasing good ones.             |
|  <b>Vitamin B5<br/>(Pantothenic Acid)</b> | Anti-stress vitamin and plays a vital role in cellular metabolism. |
|  <b>Vitamin B6<br/>(Pyridoxine)</b>       | Helps improve vision, PMS and also supports brain health.          |
|  <b>Vitamin B7<br/>(Biotin)</b>          | Promotes healthy skin, hair & nails. Improves blood sugar .        |
|  <b>Vitamin B9<br/>(Folic Acid)</b>     | Promotes healthy pregnancy & protects baby against defects.        |
|  <b>Vitamin B12<br/>(Cobalamin)</b>     | Helps make red blood cells, DNA and normal nerve function.         |

# B-COMPLEX: HEMATOPOIETIC & NEUROLOGICAL

- **Vitamin B<sub>9</sub> (Folate):**

- **Function:** One-carbon metabolism; DNA synthesis.
- **Deficiency:** Megaloblastic anemia, Neural Tube Defects (NTDs).

- **Vitamin B<sub>12</sub> (Cobalamin):**

- **Function:** Remethylation of homocysteine; isomerisation of methylmalonyl CoA.
- **Deficiency:** Pernicious anemia, subacute combined degeneration of the spinal cord.

# SOURCES OF ENERGY-RELEASING B-VITAMINS

- **B1 (Thiamine):** Whole grains, enriched cereals, legumes.
  - *Note:* Polishing rice removes the thiamine-rich bran, leading to Beriberi.
- **B2 (Riboflavin):** Milk and dairy products (easily degraded by light), eggs, and organ meats.
- **B3 (Niacin):** Unrefined and whole grains, milk, and lean meats.
  - *Note:* In corn (maize), niacin is "bound" and unavailable unless treated with alkali.
- **B5 (Pantothenic acid):** Widely distributed in all foods (Greek: *pantos* = everywhere).  
Best sources are eggs, liver, and yeast.
- **B7 (Biotin):** Liver, egg yolk, and produced by intestinal bacteria.

# SOURCES OF HEMATOPOIETIC VITAMINS

- **Vitamin B9 (Folic Acid):**

- Named from "folium" (leaf). Rich in leafy green vegetables, lima beans, and whole-grain cereals.

- **Vitamin B12 (Cobalamin):**

- **Strictly animal origin:** Liver, beef, pork, fish, eggs, and milk.
- *Clinical Pearl:* Not found in plants. Strict vegetarians (vegans) are at high risk of deficiency unless they take supplements.

# VITAMIN C (ASCORBIC ACID)

## **Functions:**

- Hydroxylation of Proline and Lysine (Collagen synthesis).
- Antioxidant properties.
- Enhances non-heme iron absorption.

**Health Significance:** \* **Scurvy:** Characterized by sore gums, loose teeth, and fragile blood vessels.

# SOURCES OF VITAMIN C

- **Primary Sources:** Citrus fruits (lemons, oranges, grapefruit), guava, amla (Indian gooseberry), tomatoes, and green peppers.
- **Sensitivity:** Vitamin C is the most heat-labile vitamin; prolonged cooking or canning destroys its activity.



# BIOCHEMICAL HEALTH SIGNIFICANCE

- Antioxidant Defense:** Vitamins A, C, and E neutralize free radicals, reducing oxidative stress and chronic disease risk.
- Metabolic Flux:** B-vitamins act as "spark plugs" for the TCA cycle and Glycolysis.
- Hormonal Action:** Vitamin D's role in bone mineral density and immune modulation.

# CLINICAL CORRELATIONS & TOXICITY

- **Hypervitaminosis A:** Increased intracranial pressure, skin desquamation.
- **Hypervitaminosis D:** Hypercalcemia and soft tissue calcification.
- **Diagnostic Markers:** \* Prothrombin time (Vitamin K).
- Erythrocyte transketolase activity (Thiamine).
- FIGLU excretion (Folate).

# CONTINUED..

- **B<sub>6</sub> (Pyridoxine):** Essential for **ALA Synthase** (rate-limiting step in Heme synthesis). Deficiency leads to Sideroblastic Anemia.
- **Biotin (B<sub>7</sub>):** Coenzyme for carboxylation reactions (e.g., Pyruvate carboxylase). Consuming raw egg whites (avidin) can cause deficiency.
- **Vitamin E Toxicity:** Least toxic of the fat-soluble group, but high doses can interfere with Vitamin K and increase the effect of anticoagulants



Thank You